

## Algorithm redundancy

Real-life representations are often prone to corruption. Biological codes, like RNA, may mutate naturally<sup>1</sup> and during measurement; cosmic radiation and other ambient noise can flip bits in computer storage<sup>2</sup>. One way to recover from corrupted data is to introduce or exploit redundancy.

Consider the following algorithm to introduce redundancy in a string of 0s and 1s.

Create redundancy by repeating each bit three times

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```

1 procedure redun3( $a_{k-1} \cdots a_0$ : a nonempty bitstring)
2 for  $i := 0$  to  $k-1$ 
3    $c_{3i} := a_i$ 
4    $c_{3i+1} := a_i$ 
5    $c_{3i+2} := a_i$ 
6 return  $c_{3k-1} \cdots c_0$ 

```

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Decode sequence of bits using majority rule on consecutive three bit sequences

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```

1 procedure decode3( $c_{3k-1} \cdots c_0$ : a nonempty bitstring whose length is an integer multiple of 3)
2 for  $i := 0$  to  $k-1$ 
3   if exactly two or three of  $c_{3i}, c_{3i+1}, c_{3i+2}$  are set to 1
4      $a_i := 1$ 
5   else
6      $a_i := 0$ 
7 return  $a_{k-1} \cdots a_0$ 

```

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Give a recursive definition of the set of outputs of the *redun3* procedure, *Out*,

Consider the message  $m = 0001$  so that the sender calculates  $\text{redun3}(m) = \text{redun3}(0001) = 000000000111$ .

Introduce \_\_\_\_ errors into the message so that the signal received by the receiver is \_\_\_\_\_ but the receiver is still able to decode the original message.

*Challenge: what is the biggest number of errors you can introduce?*

Building a circuit for lines 3-6 in *decode* procedure: given three input bits, we need to determine whether the majority is a 0 or a 1.

| $c_{3i}$ | $c_{3i+1}$ | $c_{3i+2}$ | $a_i$ | Circuit |
|----------|------------|------------|-------|---------|
| 1        | 1          | 1          |       |         |
| 1        | 1          | 0          |       |         |
| 1        | 0          | 1          |       |         |
| 1        | 0          | 0          |       |         |
| 0        | 1          | 1          |       |         |
| 0        | 1          | 0          |       |         |
| 0        | 0          | 1          |       |         |
| 0        | 0          | 0          |       |         |

<sup>1</sup>Mutations of specific RNA codons have been linked to many disorders and cancers.

<sup>2</sup>This RadioLab podcast episode goes into more detail on bit flips: <https://www.wnycstudios.org/story/bit-flip>

# Algorithm rna mutation insertion deletion

Recall that  $S$  is defined as the set of all RNA strands, nonempty strings made of the bases in  $B = \{\text{A}, \text{U}, \text{G}, \text{C}\}$ . We define the functions

$$\text{mutation} : S \times \mathbb{Z}^+ \times B \rightarrow S \qquad \text{insertion} : S \times \mathbb{Z}^+ \times B \rightarrow S$$

$$\text{deletion} : \{s \in S \mid \text{rlen}(s) > 1\} \times \mathbb{Z}^+ \rightarrow S \qquad \text{with rules}$$

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```

1 procedure mutation( $b_1 \cdots b_n$ : a RNA strand,  $k$ : a positive integer,  $b$ : an element of  $B$ )
2 for  $i := 1$  to  $n$ 
3   if  $i = k$ 
4      $c_i := b$ 
5   else
6      $c_i := b_i$ 
7 return  $c_1 \cdots c_n$  {The return value is a RNA strand made of the  $c_i$  values}

```

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```

1 procedure insertion( $b_1 \cdots b_n$ : a RNA strand,  $k$ : a positive integer,  $b$ : an element of  $B$ )
2 if  $k > n$ 
3   for  $i := 1$  to  $n$ 
4      $c_i := b_i$ 
5    $c_{n+1} := b$ 
6 else
7   for  $i := 1$  to  $k-1$ 
8      $c_i := b_i$ 
9    $c_k := b$ 
10  for  $i := k+1$  to  $n+1$ 
11     $c_i := b_{i-1}$ 
12 return  $c_1 \cdots c_{n+1}$  {The return value is a RNA strand made of the  $c_i$  values}

```

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```

1 procedure deletion( $b_1 \cdots b_n$ : a RNA strand with  $n > 1$ ,  $k$ : a positive integer)
2 if  $k > n$ 
3    $m := n$ 
4   for  $i := 1$  to  $n$ 
5      $c_i := b_i$ 
6 else
7    $m := n-1$ 
8   for  $i := 1$  to  $k-1$ 
9      $c_i := b_i$ 
10  for  $i := k$  to  $n-1$ 
11     $c_i := b_{i+1}$ 
12 return  $c_1 \cdots c_m$  {The return value is a RNA strand made of the  $c_i$  values}

```

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# Algorithm definition

**New!** An algorithm is a finite sequence of precise instructions for solving a problem.

Algorithms can be expressed in English or in more formalized descriptions like pseudocode or fully executable programs.

Sometimes, we can define algorithms whose output matches the rule for a function we already care about. Consider the (integer) logarithm function

$$\text{log}_b : \{b \in \mathbb{Z} \mid b > 1\} \times \mathbb{Z}^+ \rightarrow \mathbb{N}$$

defined by

$$\text{log}_b(b, n) = \text{greatest integer } y \text{ so that } b^y \text{ is less than or equal to } n$$

## Calculating integer part of base $b$ logarithm

```
1  procedure  $\text{log}_b(b, n$ : positive integers with  $b > 1$ )  
2     $i := 0$   
3    while  $n > b - 1$   
4       $i := i + 1$   
5       $n := n \text{ div } b$   
6    return  $i$  { $i$  holds the integer part of the base  $b$  logarithm of  $n$ }
```

Trace this algorithm with inputs  $b = 3$  and  $n = 17$

|                    | $b$ | $n$ | $i$ | $n > b - 1?$ |
|--------------------|-----|-----|-----|--------------|
| Initial value      | 3   | 17  |     |              |
| After 1 iteration  |     |     |     |              |
| After 2 iterations |     |     |     |              |
| After 3 iterations |     |     |     |              |

Compare: does the output match the rule for the (integer) logarithm function?

# Base expansion algorithms

Two algorithms for constructing base  $b$  expansion from decimal representation

**Most significant first:** Start with left-most coefficient of expansion (highest value)

*Informally:* Build up to the value we need to represent in “greedy” approach, using units determined by base.

Calculating base  $b$  expansion, from left

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```
1 procedure basebmost( $n, b$ : positive integers with  $b > 1$ )
2    $v := n$ 
3    $k := 1 + \text{output of } \log b \text{ algorithm with inputs } b \text{ and } n$ 
4   for  $i := 1$  to  $k$ 
5      $a_{k-i} := 0$ 
6     while  $v \geq b^{k-i}$ 
7        $a_{k-i} := a_{k-i} + 1$ 
8        $v := v - b^{k-i}$ 
9   return  $(a_{k-1}, \dots, a_0) \{ (a_{k-1} \dots a_0)_b \text{ is the base } b \text{ expansion of } n \}$ 
```

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**Least significant first:** Start with right-most coefficient of expansion (lowest value)

Idea: (when  $k > 1$ )

$$\begin{aligned} n &= a_{k-1}b^{k-1} + \cdots + a_1b + a_0 \\ &= b(a_{k-1}b^{k-2} + \cdots + a_1) + a_0 \end{aligned}$$

so  $a_0 = n \bmod b$  and  $a_{k-1}b^{k-2} + \cdots + a_1 = n \operatorname{div} b$ .

Calculating base  $b$  expansion, from right

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```
1 procedure basebleast( $n, b$ : positive integers with  $b > 1$ )
2    $k := 0$ 
3    $q := n$ 
4   while  $q \neq 0$ 
5      $a_k := q \bmod b$ 
6      $k := k + 1$ 
7      $q := q \operatorname{div} b$ 
8   return  $(a_{k-1}, \dots, a_0) \{(a_{k-1} \dots a_0)_b \text{ is the base } b \text{ expansion of } n\}$ 
```

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## Base conversion algorithm

Practice: write an algorithm for converting from base  $b_1$  expansion to base  $b_2$  expansion: